



Personal loans are a lucrative revenue stream for banks. The typical interest rate of a two-year loan in the United Kingdom is [around 10%](#). This might not sound like a lot, but in September 2022 alone UK consumers borrowed [around £1.5 billion](#), which would mean approximately £300 million in interest generated by banks over two years!

You have been asked to work with a bank to clean the data they collected as part of a recent marketing campaign, which aimed to get customers to take out a personal loan. They plan to conduct more marketing campaigns going forward so would like you to ensure it conforms to the specific structure and data types that they specify so that they can then use the cleaned data you provide to set up a PostgreSQL database, which will store this campaign's data and allow data from future campaigns to be easily imported.

They have supplied you with a csv file called `"bank_marketing.csv"`, which you will need to clean, reformat, and split the data, saving three final csv files. Specifically, the three files should have the names and contents as outlined below:

`client.csv`

column	data type	description	cleaning requirements
<code>client_id</code>	<code>integer</code>	Client ID	N/A
<code>age</code>	<code>integer</code>	Client's age in years	N/A
<code>job</code>	<code>object</code>	Client's type of job	Change <code>"."</code> to <code>"_"</code>
<code>marital</code>	<code>object</code>	Client's marital status	N/A
<code>education</code>	<code>object</code>	Client's level of education	Change <code>"."</code> to <code>"_"</code> and <code>"unknown"</code> to <code>np.NaN</code>
<code>credit_default</code>	<code>bool</code>	Whether the client's credit is in default	Convert to <code>boolean</code> data type: <code>1</code> if <code>"yes"</code> , otherwise <code>0</code>
<code>mortgage</code>	<code>bool</code>	Whether the client has an existing mortgage (housing loan)	Convert to <code>boolean</code> data type: <code>1</code> if <code>"yes"</code> , otherwise <code>0</code>

`campaign.csv`

column	data type	description	cleaning requirements
<code>client_id</code>	<code>integer</code>	Client ID	N/A
<code>number_contacts</code>	<code>integer</code>	Number of contact attempts to the client in the current campaign	N/A
<code>contact_duration</code>	<code>integer</code>	Last contact duration in seconds	N/A
<code>previous_campaign_contacts</code>	<code>integer</code>	Number of contact attempts to the client in the previous campaign	N/A
<code>previous_outcome</code>	<code>bool</code>	Outcome of the previous campaign	Convert to <code>boolean</code> data type: <code>1</code> if <code>"success"</code> , otherwise <code>0</code> .

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

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economics.csv

column	data type	description	cleaning requirements
client_id	integer	Client ID	N/A
cons_price_idx	float	Consumer price index (monthly indicator)	N/A
euribor_three_months	float	Euro Interbank Offered Rate (euribor) three-month rate (daily indicator)	N/A

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv("bank_marketing.csv")

for col in ["credit_default", "mortgage", "previous_outcome", "campaign_outcome"]:
    print(col)
    print("-----")
    print(df[col].value_counts())
df.head()
```

```
credit_default
-----
no          32588
unknown     8597
yes           3
Name: credit_default, dtype: int64
mortgage
-----
yes         21576
no          18622
unknown      990
Name: mortgage, dtype: int64
previous_outcome
-----
nonexistent  35563
failure      4252
success      1373
Name: previous_outcome, dtype: int64
campaign_outcome
-----
no          36548
yes          4640
Name: campaign_outcome, dtype: int64
```

...	↑↓	c.	...	↑↓	...	↑↓	j.	...	↑↓	...	↑↓	ed...	...	↑↓	credit_d...	...	↑↓	...	↑↓	...	↑↓	...	↑↓	contact_dur...	...	↑↓	nu
	0		0		56		housemaid		married		basic.4y		no		no		may		13					261			
	1		1		57		services		married		high.school		unknown		no		may		19					149			
	2		2		37		services		married		high.school		no		yes		may		23					226			
	3		3		40		admin.		married		basic.6y		no		no		may		27					151			
	4		4		56		services		married		high.school		no		no		may		3					307			

Rows: 5

Expand

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely

0

1

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Extremely likely

```
df["year"]=2022
df.head()
```

...	↑↓	c.	...	↑↓	...	↑↓	j..	...	↑↓	...	↑↓	ed...	...	↑↓	credit_d...	...	↑↓	...	↑↓	...	↑↓	...	↑↓	contact_dur...	...	↑↓	nu
	0		0		56		housemaid		married		basic.4y		no		no		may		13		261						
	1		1		57		services		married		high.school		unknown		no		may		19		149						
	2		2		37		services		married		high.school		no		yes		may		23		226						
	3		3		40		admin.		married		basic.6y		no		no		may		27		151						
	4		4		56		services		married		high.school		no		no		may		3		307						

Rows: 5

Expand

```
#splitting df into three dataframes

client=df[["client_id","age","job","marital","education","credit_default","mortgage"]]
client.head()
campaign=df[["client_id","number_contacts","contact_duration","previous_campaign_contacts","previous_outcome","campaign_outcome","day","month","year"]]
campaign.head()
economics=df[["client_id","cons_price_idx","euribor_three_months"]]
economics.head()
```

...	↑↓	c.	...	↑↓	cons_pri...	...	↑↓	euribor_three_mon...	...	↑↓
	0		0		93.994		4.857			
	1		1		93.994		4.857			
	2		2		93.994		4.857			
	3		3		93.994		4.857			
	4		4		93.994		4.857			

Rows: 5

Expand

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0

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Extremely likely

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```
#cleaning client df

client["job"]=client["job"].str.replace(".", "_")
print(client["job"])
client["education"]=client["education"].str.replace(".", "_")
print(client["education"].str.contains("unknown").sum())
client.loc[client["education"]=="unknown", "education"]=np.nan
print(client["education"].str.contains("unknown").sum())
client['credit_default'] = client['credit_default'].apply(lambda x: 1 if x=="yes" else 0)
client['mortgage'] = client['mortgage'].apply(lambda x: 1 if x=="yes" else 0)
print(client["client_id"])
client[["client_id", "age"]].astype("int64")
client["credit_default"]=client["credit_default"].astype('bool')
client["mortgage"]=client["mortgage"].astype('bool')
print(client.info())
print(client[["credit_default", "mortgage"]])
client['credit_default'].value_counts()
```

```
0      housemaid
1      services
2      services
3      admin_
4      services
...
41183    retired
41184  blue-collar
41185    retired
41186  technician
41187    retired
Name: job, Length: 41188, dtype: object
1731
0
0      0
1      1
2      2
3      3
4      4
...
41183  41183
41184  41184
41185  41185
41186  41186
41187  41187
Name: client_id, Length: 41188, dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 7 columns):
```

...	↑↓	credit_d...	...	↑↓
False		41185		
True		3		

Rows: 2

[Expand](#)

```
#cleaning campaign data
```

```
campaign['previous_outcome'] = campaign['previous_outcome'].apply(lambda x: 1 if x=="success" else 0)
campaign['campaign_outcome'] = campaign['campaign_outcome'].apply(lambda x: 1 if x=="yes" else 0)
campaign['previous_outcome']=campaign['previous_outcome'].astype('bool')
campaign['campaign_outcome']=campaign['campaign_outcome'].astype('bool')
```

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```
campaign['last_contact_date']=campaign[["year","month","day"]].astype(str).agg('-',axis=1)
campaign['last_contact_date']=pd.to_datetime(campaign['last_contact_date'])
campaign=campaign.drop(columns=["year","month","day"])
campaign.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 7 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   client_id             41188 non-null  int64  
 1   number_contacts       41188 non-null  int64  
 2   contact_duration      41188 non-null  int64  
 3   previous_campaign_contacts 41188 non-null  int64  
 4   previous_outcome      41188 non-null  bool    
 5   campaign_outcome      41188 non-null  bool    
 6   last_contact_date     41188 non-null  datetime64[ns]
dtypes: bool(2), datetime64[ns](1), int64(4)
memory usage: 1.6 MB
```

```
client.to_csv("client.csv", index=False)
campaign.to_csv("campaign.csv", index=False)
economics.to_csv("economics.csv", index=False)
```

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